Fulton Hogan Assignment

Software Modelling Report ENSE706

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1. Problem Statement

Fulton Hogan is a New Zealand based construction company that provides quality infrastructure that improves the lives of people in New Zealand and Australia. It was founded by Jules Fulton and Bob Hogan in 1933, when they teamed up with the intention of playing a part in building a prosperous country during The Great Depression period. Over the past century they have grown to over 9,000 employees (Fulton Hogan, 2024), and are known for constructing and maintaining roads, bridges, energy and water networks all across New Zealand and Australia. Their goals outside of construction have always been “creating, connecting, and caring for communities”. Fulton Hogan has several departments each responsible for an important part of the business’ operation. As of now, Fulton Hogan uses JD Edwards to manage their main operations. JD Edwards is an Enterprise Resource Planning system which manages construction projects, equipment, and payroll (Oracle, 2024). However, it is an old legacy system that is not meant to handle real-time demands of today’s construction management needs. This results in latency, data loss, and errors during processing in all the departments.

Fulton Hogan is looking to improve their Project Costing & Management System, which involves three departments; Project Management department, Finance department, and Operations department. Each department has three main responsibilities which allow Fulton Hogan to run as a successful business.

1.1 Project Management Department

The main responsibilities of this department are:

- Creating and managing projects
- Logs timesheets, milestones, and project progress
- Submits budget requests and cost updates

The Project Management department serves as the primary driver of project planning and execution. They are responsible for the entire process of a construction project from start to finish. This department ensures projects are on schedule, and ensures that any tasks are completed within the time given.

The main users of this department are Project Managers, Site Leads and Project Coordinators. Project Managers are the people that manage the entire project. When given a project, they are responsible for managing both people and resources. Site

Leads are the people who are on the construction sites and oversee everything going on, making sure everything is following the correct procedure. Project Coordinators are the people working on the project and contributing their input. They are always finding solutions to any problems that arise, making sure everything is working smoothly, and acting as a central point of communication.

The Project Management Department uses the system as the central hub for data entry and project tracking. They initialise projects, log any progress made, and any updates are also triggered by these three user roles. Each user has different privileges which allow them to access different parts of the system. Project Managers can initialise projects and add any major changes to them (extending deadlines, generating team reports, managing the team). However, they cannot see what the Site Leads are telling the Site Foremen. In the new system, they should be able to check in and make sure the goals of the project are being expressed. Site Leads use the system to guide general labourers and construction workers on-site. They need all information available to them and the tasks that are needed to achieve in a selected time period. Project Coordinators use the system to document any new findings, managing new information, analysing data, and creating reports for the Project Manager.

All users of this system have reported latency issues. Whenever Project Coordinators contribute to the project, Project Managers cannot see real-time progress meaning the Site Leads don't know what to do in current time. This slows down progress because the Site Lead has to wait before giving out instructions to the workers. This is due to the outdated legacy program, causing scheduling conflicts and data inconsistences.

1.2 Finance Department

The main responsibilities of this department are:

- Approving and monitoring project budgets
- Tracking actual vs estimated costs
- Processes financial reporting on projects

The Finance department is responsible for overseeing the financial aspects of project delivery. They ensure that the company is always making profits by reviewing every dollar allocated to the project. This department is always focused on financial data and making sure everything on the financial side is going smoothly.

Financial Controllers, Project Accountants, and Auditors are users for this department. Financial controllers manage the day-to-day financial operations such as monthly reporting, managing cash flow, and ensuring accounting standards are

met across project timelines. Project Accountants have the main responsibility of handling financial tracking for projects. Auditors are always reviewing financial records and processes to verify financial accuracy when it comes to project resources.

The Finance Department uses the system as the primary platform for financial analysis and reporting, making sure all expenses remain aligned with approved budgets. They are always comparing Estimated vs Actual (EVA) costs, approving the funds for Project Management requests, and generating monthly compliance reports. Similar to the Project Management Department, the users in this department have different privileges on the system. The Financial Controllers have access to all resources so that they can make profitable decisions. Project Accountants have access to tracking software, however not to the reports that show project spendings. They should have access to these in the new system. Auditors have access to all financial reports, because they need to have all financial information to create reports for the Project Managers.

All users have reported latency issues regarding system updates due to the old legacy system. They need to always wait before seeing important updates that should be available instantaneously. Auditors have recently reported missing data every now and then. The Project Accountants remember entering the data, however the Auditors cannot see them in the reports, meaning that data has been lost during processing. Fulton Hogan wants to put priority into this part of the system since it allows decisions to be made which leads the project progress forwards.

1.3 Operations Department

The main responsibilities of this department are:

- Executing the on-ground construction work
- Reporting labour hours and equipment usage
- Flagging project issues that affect cost or timeline

The Operations department is responsible for handling all the physical execution, heavy machinery, labour and materials. Their responsibility is to safely and efficiently deliver the infrastructure for their clients.

The main users of this department are the Site Foremen, Heavy Equipment Operators, and General Laborers. The Site Foremen are the supervisors of this department, always managing the day-to-day activities of all the workers. They are constantly translating instructions given by the Site Leads, and other higher ups. Heavy Equipment Operators are in charge of operating all the tools, vehicles, and equipment used to deliver the infrastructure. They use the system for communicating

with the Site Foremen and General Laborers, and report anything to do with the equipment and machines. However they should also have access to the instructions given by the Site Lead. This will ensure they are prepared for the upcoming tasks that day and can plan ahead. The General Laborers are the entry level workers who carry out physical tasks on site. They use the system for communication, reporting any problems, and logging in their hours.

This department uses the system as a daily activity log, where everything that happened in the work day is reported and given to people back at the company. The data reported by this department include how many people were working, state of the machines, work to be done the next day, etc. They also have a channel dedicated to reporting issues e.g. broken vehicle or machine.

The users from this system have consistently reported performance issues with JD Edwards. Site Foremen are always waiting for instructions to come through, and progress is slowed down until it updates. Workers have reported recurring faults in their paychecks. Even when putting in the correct number of hours worked, the system doesn't process it properly. Fulton Hogan wants to find the system part responsible for this and remake it, so that it can handle the demands of thousands of employees a day.

2. System Overview

Fulton Hogan wants to develop a completely new and updated Project Costing & Management System that allows all three departments to function effortlessly and efficiently. The system should allow any data inputted to go through successfully and without issue, at the same time the other devices using this software should be updated instantaneously. All users should have the correct amount of access, while some users get more privileges than others depending on their role in the company. Others should have updated privileges from the old system, and have more access to important documents that will help them work more effectively. The new Project Costing & Management system should improve efficiency, reduce the time it takes for others to be notified of changes to any data or projects, and provide a much better experience to all system users. This will be done by applying modern software principles that can meet the demands of modern project management.

2.1 System Scope

The new Project Costing & Management System should improve real-time data entry, tracking, and reporting across all 3 departments at Fulton Hogan: Project

Management, Finance, Operations. Everything mentioned about the departments above will be key features in the scope: Project initialisation and management, timesheet logging, budget management, cost tracking, financial reporting, equipment reporting, issue flagging for equipment, and role-based privileges for the different users of the system.

2.2 System Boundaries

This system is limited to the internal operations of Fulton Hogan's Project Costing & Management function. It will not replace or directly integrate with any other external systems that are used by Fulton Hogan or their users. Human Resources, recruitment, and employee reviews are examples of functions that fall out of the scope of this system. The only functions that are being improved are the Project Management department, Finance department, and Operations department. All other business functions fall outside the scope of this system, and are managed by other software. This was confirmed with the client through consultation during the first meeting. Based on the consultation, it has been determined that all requirements have been captured within the scope. There are no other missing requirements.

2.3 Assumptions

The following assumptions have been made during the analysis of this system.

All users of this new system will have access to a computer (personal or company provided), along with a stable internet connection. This is needed because the system has real-time operations that require a strong connection to see the new updated information.
Fulton Hogan will provide detailed training to all of their employees (managers, labourers, office workers) to switch over to the new system, once it has passed testing and has been deployed.
Each user will have an account that is registered to them from JD Edwards. All of their data will go through to the new system and be securely stored, so that new accounts do not have to be made. On top of this, there will be a requirement for the user to have a Fulton Hogan specific email, as any other email will be blocked from logging into the new system.
Each user role will have their corresponding privileges within the system, these privileges are described in the Problem Statement. Users in the Project Management department will have exclusive access to tools linked with project management. Users in the Finance department will have access to various financial tools, allowing them to analyse financial data. Users in the Operations department will have a timesheet subsystem, along with a flagging issues page, and more.

Existing data stored in JD Edwards will be migrated to the new system without any significant losses. It should be noted that Fulton Hogan is committed to maintaining data integrity during the transition period.

3. Requirements Engineering

This section defines both functional and non-functional requirements of the new Project Costing & Management System for Fulton Hogan.

Functional requirements describe what the system must do, these are important for the functionality of the system. Non-functional requirements describe the quality standards the system must meet. This could be performance related like update speeds, or it could be related to security and reliability.

3.1 Functional Requirements

Project Management Department:

Number	Description	Stakeholder
01	The system shall allow Project Managers to initialise a new project.	Project Manager
02	The system shall allow Project Managers to modify project details.	Project Manager
03	The system shall allow Project Managers to manage team members.	Project Manager
04	The system shall allow Project Managers to view real-time progress updates from all team members.	Project Manager
05	The system shall allow Project Coordinators to log progress, findings, and data entries.	Project Coordinator
06	The system shall allow Project Coordinators to generate reports of their work.	Project Coordinator
07	The system shall allow Site Leads to view all project tasks and instructions in real time.	Site Lead
08	The system shall allow Site Leads to	Site Lead

	communicate instructions directly to Site Foremen.	

Finance Department:

Number	Description	Stakeholder
01	The system shall allow Financial Controllers to approve all project budgets.	Financial Controller
02	The system shall allow Financial Controllers to access all financial resources.	Financial Controller
03	The system shall allow Project Accountants to access financial tracking tools.	Project Accountants
04	The system shall allow Project Accountants to access project spending reports.	Project Accountants
05	The system shall allow Auditors to access all financial reports for verification.	Auditors
06	The system shall generate monthly financial reports for the Finance Department	Financial Department
07	The system shall track and compare estimated versus actual project costs in real time.	Financial Department

Operations Department:

Number	Description	Stakeholder
01	The system shall allow Site Foremen to receive and view instructions from Site Leads in real time.	Site Foremen
02	The system shall allow Site Foremen to coordinate on-site activities.	Site Foremen
03	The system shall allow Heavy Equipment Operators to log equipment usage and condition	Heavy Equipment Operators
04	The system shall allow Heavy Equipment Operators to view instructions from Site Leads	Heavy Equipment Operators
05	The system shall allow General Labourers to log daily working hours.	General Labourers

06	The system shall allow General Labourers to report on-site issues through the system.	General Labourers
07	The system shall provide a dedicated issue flagging channel for reporting equipment faults.	Operations Department

3.2 Non-Functional Requirements

Performance:

Number	Description	Stakeholder
01	The system shall synchronise data across all departments in under 2 seconds.	All Departments
02	The system shall support a minimum of 2000 concurrent users without the performance degrading.	All Users
03	The system shall process and display timesheet entries within 3 seconds of submission.	Operations Department

Reliability:

Number	Description	Stakeholder
01	The system shall maintain a minimum uptime of 99.9% during business hours.	All Departments
02	The system shall prevent data loss during processing and transmission between departments.	Finance Department
03	The system shall maintain a full audit log of all data entries and modifications.	Finance Department
04	The system shall accurately process and record all submitted labour hours without data loss.	Operations Department

Security:

Number	Description	Stakeholder
01	The system shall encrypt all data transmitted between users and server to hide information from outsiders.	All Users
02	The system shall restrict access to system features based on the user's role in the company.	All Users
03	The system shall only allow users with a verified Fulton Hogan email address to sign into the system.	All Users
04	The system shall comply with the New Zealand Privacy Act 2020.	All Users

References

Fulton Hogan. (2024). Retrieved from https://www.fultonhogan.com/

Oracle. (2024). JD Edwards EnterpriseOne, Retrieved from https://www.oracle.com/anz/applications/jd-edwards-enterpriseone/#rc30p4